Economic Observability (Non-Normative)

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ABSTRACT

We introduce a non-normative formulation for observing and comparing the economic footprint of AI-mediated software construction across specification graph nodes and time windows. The goal is diagnostic: to expose inference, verification, and ownership proxies as observable quantities under a chosen pricing surface, enabling longitudinal and comparative analysis rather than optimization prescriptions.

Keywords: software engineering — AI-assisted programming — software metrics — observability — cost modeling

1. ECONOMIC OBSERVABILITY

1.1. *Pricing Surface and Footprints*

Let each specification node be denoted by $u \in V$, and let t denote an iteration window (e.g., a generation run, a PR cycle, or a bounded time slice).

We assume a configurable *pricing surface* for the active toolchain:

- $\mathcal{M}$ — the set of AI models available in the product.
- $\mathcal{J}$ — the set of external tools (search, build, test, OCR, etc.).
- p_m — price per token (or billable token unit) for model $m \in \mathcal{M}$.
- q_m — fixed per-call price for model m (optional; $q_m = 0$ if none).
- r_j — price per invocation for tool $j \in \mathcal{J}$ (optional).

For each node u and window t, we observe the footprint tuple

$$\big(T_m(u,t),\, N_m(u,t)\big)_{m\in\mathcal{M}}\,, \quad \big(N_{\mathrm{tool},j}(u,t)\big)_{j\in\mathcal{J}}\,,$$

where $T_m(u,t)$ is the (billable) token volume under model m, $N_m(u,t)$ is the number of model calls, and $N_{\mathrm{tool},j}(u,t)$ is the number of invocations of external tool j.

1.2. *Node Inference Cost*

We define the *node inference cost* as the pricing-functional applied to the observed footprint:

$$C_{\mathrm{inf}}(u,t) \equiv \sum_{m\in\mathcal{M}} \Big(p_m\, T_m(u,t) + q_m\, N_m(u,t)\Big) \;+\; \sum_{j\in\mathcal{J}} r_j\, N_{\mathrm{tool},j}(u,t). \tag{1}$$

This value is directly displayable on the map as currency (\$, €, etc.) under a fixed pricing surface.

1.3. *Graph (Subgraph) Inference Cost*

For any subgraph or selection $S \subseteq V$ (e.g., a feature branch, epic, release slice), we define:

$$C_{\mathrm{inf}}(S,t) \equiv \sum_{u\in S} C_{\mathrm{inf}}(u,t). \tag{2}$$

1.4. *Differential Economics*

Economic observability is intended to be interpreted comparatively. We define the discrete delta and drift-velocity operators for inference cost:

$$\Delta C_{\mathrm{inf}}(u;t_k) \equiv C_{\mathrm{inf}}(u,t_k) - C_{\mathrm{inf}}(u,t_{k-1}), \tag{3}$$

$$\mathrm{DV}_C(u,t) \equiv \frac{d\,C_{\mathrm{inf}}(u,t)}{dt}, \qquad \mathrm{DV}_C(u,t_k) \approx \frac{C_{\mathrm{inf}}(u,t_k) - C_{\mathrm{inf}}(u,t_{k-1})}{t_k - t_{k-1}}. \tag{4}$$

Analogous operators apply to any node- or graph-level metric.

1.5. *Verification Cost*

We define a separate (configurable) verification cost functional:

$$C_{\mathrm{verify}}(u,t) \equiv \sum_{\ell \in \mathcal{L}} s_\ell \, N_{\mathrm{verify},\ell}(u,t), \tag{5}$$

where $\mathcal{L}$ is the set of verifiers (CI workflows, device farm runs, property test suites, performance regressions, etc.), $N_{\mathrm{verify},\ell}(u,t)$ is the number of verification runs of type ℓ, and s_ℓ is the unit price for ℓ (or an internal cost proxy).

1.6. *Total Build Cost*

We define the *build cost* for a node or selection as:

$$C_{\mathrm{build}}(u,t) \equiv C_{\mathrm{inf}}(u,t) + C_{\mathrm{verify}}(u,t), \qquad C_{\mathrm{build}}(S,t) \equiv \sum_{u \in S} C_{\mathrm{build}}(u,t). \tag{6}$$

C_{build} is the direct economic footprint of producing a concrete artifact state from intent under the current toolchain configuration.

1.7. *Change-Cost Proxy (Graph-Based)*

We introduce a non-normative *change-cost proxy* to quantify how costly future modifications may become due to graph coupling and instability. Let $\mathrm{WCR}_k(u)$ denote the weighted change radius within hop distance k, and let $\mathrm{SVC}(u)$ and $\mathrm{CSI}(u)$ denote intent verifiability and checkpoint stability, respectively (defined elsewhere).

$$C_{\mathrm{change}}(u) \equiv \kappa \, \mathrm{WCR}_k(u) \, \frac{1}{\mathrm{SVC}(u) + \varepsilon} \, \frac{1}{\mathrm{CSI}(u) + \varepsilon}, \tag{7}$$

where κ is a project-calibration coefficient and $\varepsilon > 0$ stabilizes the expression. This quantity is diagnostic: its meaning lies in relative comparisons and longitudinal drift.

1.8. *Total Ownership Proxy*

Finally, we define a composite *ownership proxy* over a selection S:

$$C_{\mathrm{own}}(S) \equiv C_{\mathrm{build}}(S) + \lambda \sum_{u \in S} C_{\mathrm{change}}(u), \tag{8}$$

where λ encodes the chosen support horizon (how much weight is assigned to future change cost). C_{own} is not an estimate guarantee; it is an economic lens for comparing structural alternatives and observing drift under AI-mediated construction.

1.9. *Interpretation Note*

All economic quantities are non-normative observables. Absolute values depend on pricing, tooling, and policy configuration. The recommended mode of use is comparative: deltas (Δ), drift velocities (DV), distribution shifts across the graph, and cost bands under controlled configurations.

1.10. *Map View Encoding (UI Mapping)*

We treat the map as a *projection* of observability signals onto visual channels. Let $x(u) \in \mathbb{R}^2$ denote the fixed layout coordinate of node u in the map view. Under a selected scope $S \subseteq V$ and time window t, the UI encodes:

- **Fill color (economic heat):** node fill is mapped to instantaneous inference cost

$$\mathrm{Color}(u) \Leftarrow C_{\mathrm{inf}}(u,t), \tag{9}$$

  typically via a monotone colormap with optional banding.

- **Node radius (growth / pressure):** node size reflects a chosen differential economic signal, by default

$$\mathrm{Radius}(u) \Leftarrow |\Delta C_{\mathrm{inf}}(u;t_k)| \quad \text{or} \quad \mathrm{Radius}(u) \Leftarrow |\mathrm{DV}_C(u,t)|\,. \tag{10}$$

- **Halo / influence ring (blast radius):** an outer ring visualizes change impact at hop distance k,

$$\mathrm{Halo}(u) \Leftarrow \mathrm{WCR}_k(u), \tag{11}$$

  where the ring radius or intensity increases with weighted reachability.

- **Border style (confidence / stability):** border opacity or dash pattern encodes verification and stability,

$$\mathrm{Border}(u) \Leftarrow \big(\mathrm{SVC}(u),\, \mathrm{CSI}(u,t)\big), \tag{12}$$

  where low SVC (weak verifiability) and low CSI (unstable checkpoints) are rendered with reduced opacity and/or higher visual "noise".

- **Edge emphasis (coupling cost flow):** when edges are shown, thickness or glow corresponds to the marginal cost transfer associated with coupling, e.g.

$$\mathrm{EdgeWeight}(u,v) \Leftarrow \Delta C_{\mathrm{inf}}(u;t_k)\,\mathbf{1}_{(u\to v)} \quad \text{or} \quad w(u,v), \tag{13}$$

  where $w(u,v)$ is an application-defined coupling weight (dependency strength, co-change frequency, or interface-touch severity).

- **Scope totals (header KPI, non-normative):** the current selection S reports aggregated economics

$$\mathrm{Header}(S) \Leftarrow \Big(C_{\mathrm{inf}}(S,t),\, C_{\mathrm{build}}(S,t),\, C_{\mathrm{own}}(S)\Big), \tag{14}$$

  always accompanied by the active pricing surface and time window.

The mapping is explicitly *lens-based*: the user may choose which economic signal drives each channel (color, radius, halo, border), while the system preserves comparability by fixing the pricing surface and time window for a given analysis session.

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- **Halo / influence ring (blast radius):** an outer ring visualizes change impact at hop distance k,
$$\mathrm{Halo}(u) \Leftarrow \mathrm{WCR}_k(u), \tag{17}$$
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$$\mathrm{Border}(u) \Leftarrow \big(\mathrm{SVC}(u),\, \mathrm{CSI}(u,t)\big), \tag{18}$$
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where $w(u,v)$ is an application-defined coupling weight (dependency strength, co-change frequency, or interface-touch severity).

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$$\mathrm{Header}(S) \Leftarrow \Big(C_{\mathrm{inf}}(S,t),\, C_{\mathrm{build}}(S,t),\, C_{\mathrm{own}}(S)\Big), \tag{20}$$
always accompanied by the active pricing surface and time window.

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